NEW STAFF E-LEARNING PROGRAM

Facility Management & Safety

Aligned to CBAHI accreditation standards, MOH emergency codes, and ISO 45001

1. CBAHI FMS Chapter Overview
2. Occupational Health & Safety
3. Disaster Management
4. Fire Safety Plan
5. Hazardous Materials & Waste Management
6. Security Management Plan
7. Utility Plan
8. Biomedical Plan
9. Radiation & Laser Safety

[Facility Name] · Safety & Security Department · [Month Year]

MODULE 1 OF 9

CBAHI Facility Management & Safety Chapter Overview

By the end of this module, you will be able to name each of the eight Facility Management & Safety plans and describe what each one exists to do.



Fire · Building · Hazmat & Waste · Utility · Biomedical · Security · Internal Disaster · External Disaster

FRAMING

Why This Chapter Exists

CBAHI accreditation requires every healthcare facility to demonstrate that its physical environment, equipment, and emergency systems actively protect patients, staff, and visitors. Facility Management & Safety is not a back-office formality — it is a surveyed accreditation chapter, built on eight interlocking plans that together cover every physical risk in the building.



A survey requirement

CBAHI surveyors assess each of these eight plans directly — documentation, staff knowledge, and physical evidence all matter.



Everyone's responsibility

These plans only work if every employee — clinical or non-clinical — knows their role in each one.



One connected system

The eight plans overlap in practice: a fire is a Code Red, a chemical spill is a Code Orange — the plans and the emergency codes are two views of the same system.

THE FRAMEWORK

Eight Plans, One Chapter



Fire Safety



Building Safety



Hazmat & Waste



Utility



Biomedical



Security



Internal Disaster



External Disaster

PLAN DETAIL

Fire Safety Plan & Building Safety Plan

Fire Safety Plan

PURPOSE

Protect people from harm caused by fire and smoke.

MAIN FUNCTIONS

- Prevention & routine inspections
- Detection & suppression systems
- Code Red response (RACE / PASS)
- Evacuation routes & drills

Building Safety Plan

PURPOSE

Keep the physical environment free of hazards.

MAIN FUNCTIONS

- Environmental & safety rounds
- Hazard identification & correction
- Construction/renovation risk assessment
- Safety signage & wayfinding

PLAN DETAIL

Hazmat & Waste Management Plan & Utility Plan



Hazmat & Waste Management Plan

PURPOSE

Safely manage chemicals and medical waste end-to-end.

MAIN FUNCTIONS

- Inventory, SDS & GHS/NFPA labeling
- Safe storage, handling & transport
- Code Orange spill response
- Waste segregation & disposal



Utility Plan

PURPOSE

Keep essential utility systems running safely.

MAIN FUNCTIONS

- Critical system inventory
- Preventive maintenance program
- Backup power & water continuity
- Code Brown response

PLAN DETAIL

Biomedical Plan & Security Plan



Biomedical Plan

PURPOSE

Ensure medical equipment is safe and functional.

MAIN FUNCTIONS

- Equipment inventory & inspection
- Planned preventive maintenance (PPM)
- Incident & recall management
- User training on equipment



Security Plan

PURPOSE

Protect people, property, and information.

MAIN FUNCTIONS

- Access control & ID badging
- Visitor & contractor management
- CCTV monitoring
- Response to Codes White/Silver/Black/Pink

PLAN DETAIL

Internal Disaster Plan & External Disaster Plan



Internal Disaster Plan

PURPOSE

Prepare for emergencies originating inside the facility.

MAIN FUNCTIONS

- Emergency code activation protocols
- Emergency Response Team (ERT) roles
- Continuity of critical services
- Regular internal drills



External Disaster Plan

PURPOSE

Prepare for mass-casualty or community events affecting the facility.

MAIN FUNCTIONS

- Surge capacity planning
- Coordination with Civil Defense / MOH / Red Crescent
- Code Yellow triage & communication
- Community disaster drills

MODULE 2 OF 9

Occupational Health & Safety

By the end of this module, you will understand core OHS definitions, how to identify and control risk, who to contact in your department, and your rights under ISO 45001.



Health & Safety · Hazards · Risk · Your Rights (ISO 45001)

Core Definitions



Health

A state of complete physical, mental, and social well-being — not merely the absence of injury or illness.



Safety

Freedom from conditions that can cause injury, illness, or damage — achieved by controlling hazards before they cause harm.



Hazard

Anything with the potential to cause harm — a wet floor, an unlabeled chemical, an exposed wire, a violent individual.



Risk

The likelihood that a hazard will actually cause harm, combined with how severe that harm would be.

KNOW THE CATEGORIES

Incident Types



Near miss

Nothing happened, but it easily could have — the single most valuable incident type to report.



First aid case

Minor injury treated with on-site first aid; no further medical care required.



Medical treatment case

Injury requiring treatment beyond first aid, such as stitches or prescribed medication.



Lost time injury

Injury serious enough that the employee cannot return to their normal duties for a period of time.



Fatality

A death resulting from a workplace incident — triggers mandatory investigation and regulatory reporting.

The AREC Framework



Anticipate

Think ahead about what could go wrong before you start a task or enter an area.



Recognize

Identify hazards as they actually appear in your environment or task.



Evaluate

Assess how likely the hazard is to cause harm and how severe that harm would be.



Control

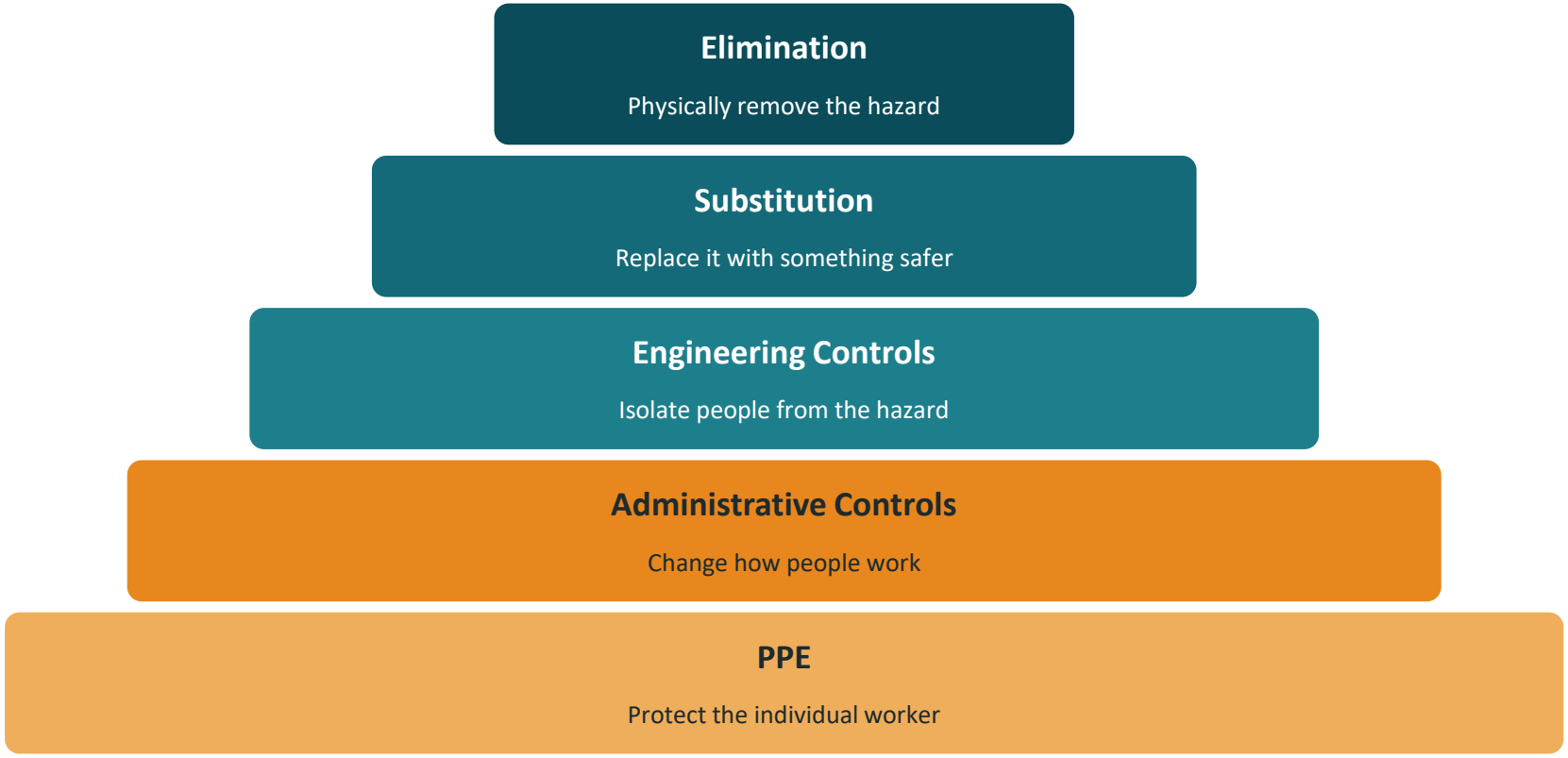
Apply the Hierarchy of Controls to reduce or remove the risk, starting with the most effective option.

This cycle repeats continuously — hazard awareness is never a one-time task.

RISK CONTROL

Hierarchy of Controls

Start at the top. The higher a control sits, the more reliably it removes the risk — PPE is the last line of defense, not the first.



WHO TO ASK

Department Safety Liaisons

Department Safety Liaisons are appointed by their departments to facilitate safety information and comply with the Environmental Health & Safety (EHS) department.



Ask your supervisor

If you don't know who your department's Safety Liaison is, your direct supervisor can tell you on your first day.



What they do

They're your first point of contact for hazard reporting, risk assessments, and PPE questions — and the bridge to the EHS department.

KNOW YOUR RIGHTS

Your Rights Under ISO 45001

These rights exist so that safety concerns are never silenced by fear of consequences.



The right to stop unsafe work

You have the right to stop any task you believe could cause serious harm — without fear of retaliation.



The right to be consulted

If you're afraid to stop work yourself, you have the right to be consulted and to raise the concern with your Safety Officer or supervisor.



The right to training

You have the right to be trained on the hazards and safe procedures relevant to your role before you're expected to perform it.

MODULE 3 OF 9

Disaster Management

By the end of this module, you will recognize all MOH-approved emergency codes, know when and how each is activated, and understand the role of the Emergency Response Team.



Fire · Hazmat · Security · Medical · Weather · Utility · External Disaster

KNOW BEFORE YOU NEED IT

Emergency Codes (1 of 2)

Announced facility-wide over the PA system — every code below is approved by the Ministry of Health.

Code Red

Fire

Code Orange

Hazardous material spillage

Code White

Violent situation

Code Black

Bomb threat

Code Pink

Child/infant abduction or lost

Code Blue

2222

Cardiac arrest

KNOW BEFORE YOU NEED IT

Emergency Codes (2 of 2)

Code RRT

Deteriorating patient (Rapid Response Team)

Code Silver

Violent situation involving a weapon

Code Yellow

External disaster

Code Grey

Severe weather

Code Brown

Utility failure

These 11 codes are this facility's approved MOH set. Extension numbers vary by site — 2222 (Code Blue) is standardized nationally by the Ministry of Health.

Facility-Wide Announcements

Hospital Evacuation

A distinct, facility-wide announcement directing the full evacuation of the building — follow your unit's evacuation plan and proceed to the outdoor assembly point immediately.

End of Disaster

Announced once the Incident Commander confirms the emergency is fully resolved — this is your signal that it's safe to stand down and resume normal operations.

THE COMMON PATTERN

General Response Framework



Recognize

Notice a hazard or emergency situation as it develops.



Activate

Call the switchboard/security and announce the correct code overhead.



Respond

Follow your role-specific response — every staff member has a part to play.



Stand Down

Wait for the official all-clear before resuming normal operations.

The specific actions differ by code, but this sequence applies to all of them.

The Emergency Response Team (ERT)



Safety



Security



Nursing



Facility



Operation

MODULE 4 OF 9

Fire Safety Plan

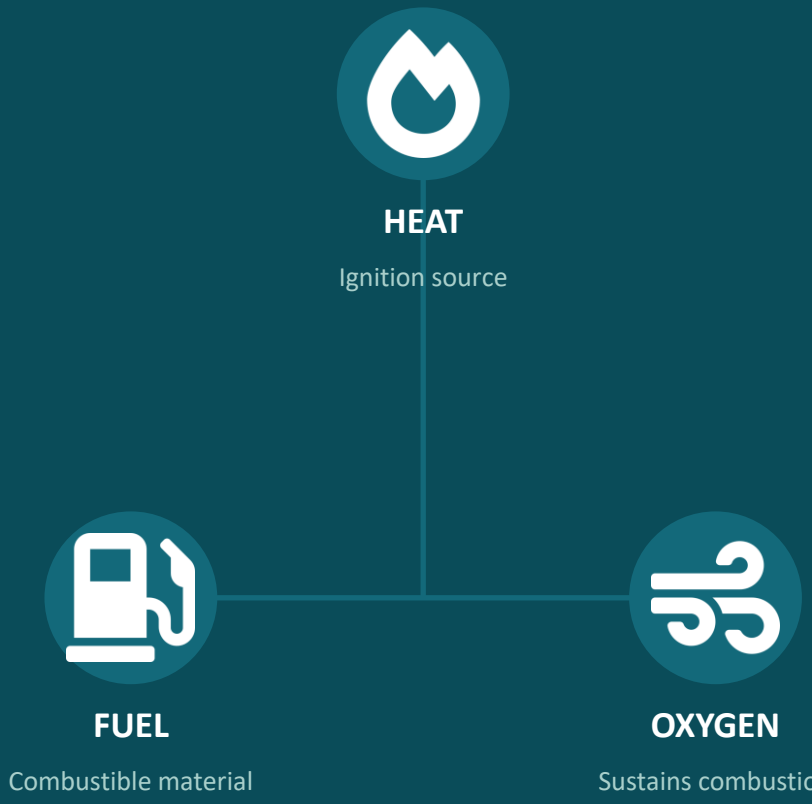
By the end of this module, you will understand how fire behaves, how to respond using RACE and PASS, and what fire prevention rules apply to everyone.



Fire Triangle · RACE / PASS · Evacuation · Prohibited Items

The Fire Triangle

Remove any one of these three elements and the fire cannot start or continue — this is the logic behind every fire control method.



Fire Behavior

A fire starts small and grows exponentially as heat radiates outward, igniting nearby fuel. Smoke — not flame — is the leading cause of death in facility fires, spreading faster than fire through corridors and duct systems.

This is why RACE prioritizes rescue and containment before extinguishing.

Fire Response: RACE & PASS

RACE — if you discover a fire

1. **Rescue**
   Move anyone in immediate danger away from the fire.
2. **Alarm**
   Activate the nearest fire alarm pull station and call switchboard/security — state the code type, your location, and your ID & name.
3. **Contain**
   Remove flammables like oxygen cylinders and alcohol-based sanitizers, then close doors and windows to slow the spread of smoke and fire.
4. **Extinguish or Evacuate**
   Use an extinguisher only for a small, contained fire — otherwise evacuate by the nearest safe route.

PASS — using a fire extinguisher

1. **Pull**
   Pull the safety pin at the top of the extinguisher.
2. **Aim**
   Aim the nozzle low, at the base of the fire.
3. **Squeeze**
   Squeeze the handle to release the extinguishing agent.
4. **Sweep**
   Sweep side to side until the fire is fully out.

KNOW YOUR EQUIPMENT

Fire Alarm Equipment



Smoke Detector



Heat Detector



Call Point

KNOW YOUR EQUIPMENT

Firefighting & Evacuation Equipment



Fire Extinguishers



Fire Blanket



Sprinkler System



Evacuation Routes



Evacuation Chairs



Evacuation Mattress



Baby Mover / Apron



Assembly Points

MATCH THE EXTINGUISHER TO THE FIRE

Fire Extinguisher Agents



CO2

Electrical fires



ABC Powder

General-purpose



Water Spray

Wood & paper



Foam Spray

Flammable liquids



Wet Chemical

Cooking oil fires

KNOW YOUR CLASSES

Fire Extinguisher Classes

Using the wrong extinguisher class can make a fire worse — check the label before you pull the pin.

Class A

Wood, paper, fabric, trash

Class B

Flammable liquids & gases

Class C

Energized electrical equipment

Class D

Combustible or burning metals

Class K

Cooking oils & fats

MATCH THE RESPONSE TO THE FIRE

Types of Evacuation



Horizontal

Move patients and staff through fire/smoke doors to an adjacent safe compartment on the same floor — the first response for most fires.



Vertical

Move down to a lower floor or ground level using stairwells — never elevators — when horizontal evacuation isn't sufficient.



Full Evacuation

Evacuate the entire building to the outdoor assembly point — reserved for the most severe, facility-wide emergencies.

What Not to Bring or Use

- Candles & Open Flame
- Unauthorized Welding
- Unauthorized Extensions
- Heaters & Electrical Boilers
- Indoor Smoking & Vaping
- Unauthorized Microwaves
- Uncontrolled Flammables

IMPORTANT CLARIFICATION

Oxygen is not flammable — it is an oxidizer.

Oxygen doesn't burn on its own, but it dramatically accelerates combustion and makes existing fires burn hotter and faster. Never allow oxygen cylinders or lines near open flame, sparks, or smoking materials — the risk isn't ignition, it's intensification.

MODULE 5 OF 9

Hazardous Materials & Waste Management

By the end of this module, you will know how chemicals are ordered, labeled, stored, and handled — how to respond to Code Orange — and how to segregate medical waste correctly.



Chemicals · Labels · Storage & Transport · Spill Response · Waste

BEFORE IT ARRIVES ON SITE

Chemical Ordering & Inventory



Approved ordering process

Chemicals are ordered only through the approved procurement process — no personal or unlogged purchases of hazardous substances.



Inventory & SDS

Every chemical in your department must appear on the hazardous materials inventory with its Safety Data Sheet (SDS) accessible.



GHS labeling

Globally Harmonized System (GHS) pictograms on every container show the hazard class at a glance — never accept an unlabeled container.

READ BEFORE YOU USE

GHS Pictograms

Official format: a black hazard symbol on a white background, framed by a red diamond border.



Skull & Crossbones

Acute toxicity



Corrosion

Skin & eye damage



Exploding Bomb

Explosives



Flame

Flammable



Exclamation Mark

Irritant



Health Hazard

Carcinogen / toxicity



Gas Cylinder

Gases under pressure



Flame Over Circle

Oxidizers

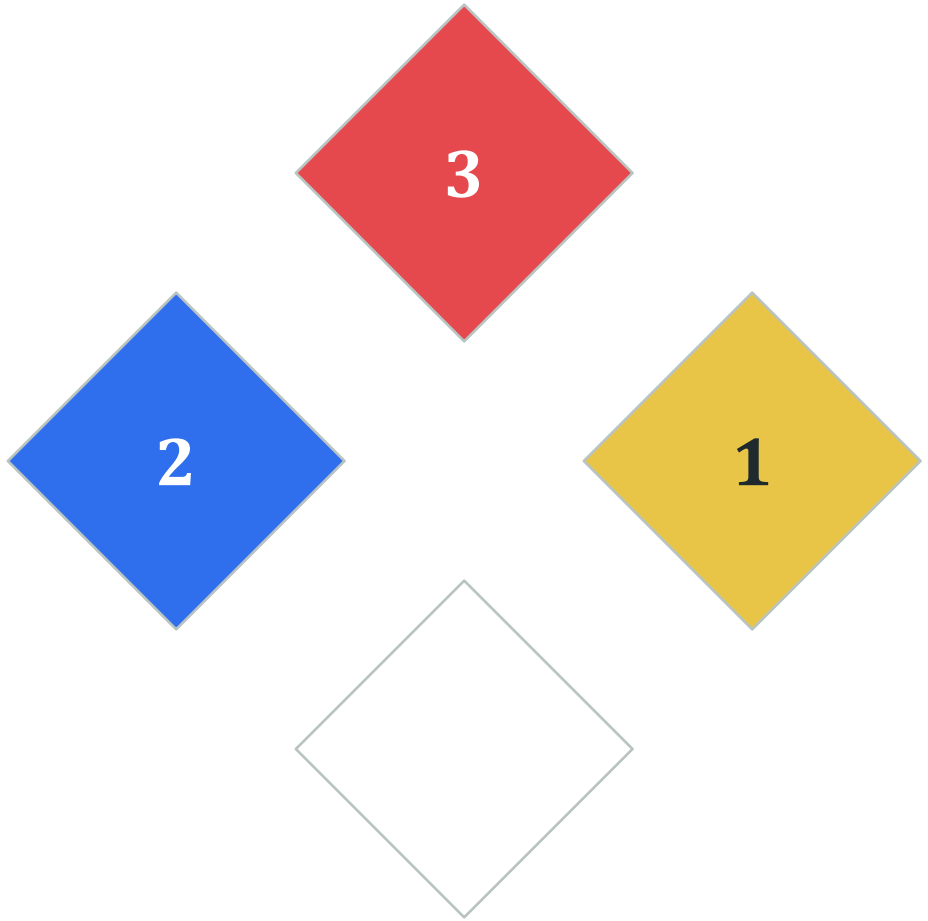


Environment

Aquatic toxicity

Reading the NFPA 704 Diamond

Every hazardous chemical container should carry this diamond. Each colored section rates a different hazard from 0 (minimal) to 4 (severe).



Blue — Health

0 = no hazard, 4 = deadly

Red — Flammability

0 = will not burn, 4 = extremely flammable

Yellow — Reactivity

0 = stable, 4 = may detonate

White — Special

Symbols like OX (oxidizer), W (avoid water)

READING THE NUMBERS

NFPA Rating Scale (0–4)

Higher numbers mean greater hazard — always check the diamond before you open, mix, or store a chemical.



Blue — Health

0 = no hazard → 4 = can be lethal with minimal exposure.



Red — Flammability

0 = will not burn → 4 = will vaporize and readily burn at normal temperature.



Yellow — Reactivity

0 = stable → 4 = may detonate at normal temperature or pressure.



White — Special hazard

Symbols only — e.g. OX (oxidizer), W (avoid water), COR (corrosive), ALK (alkaline).

Storage, Handling & Transport



Storage

Store chemicals per their SDS — segregated from incompatible substances, secured, and away from heat sources.



Transport, loading & unloading

Use designated carts and routes; never carry incompatible chemicals together, and secure containers before moving them.



Handling

Wear the PPE specified on the SDS before opening, mixing, or dispensing any chemical.

BUILT FOR CONTAINMENT

Safety Cabinets & SDS Access



Flammable storage cabinets

Double-walled, welded steel cabinets with adjustable shelves and a self-latching door — keep flammable liquids in an approved cabinet, never on an open shelf.



SDS at point of use

A Safety Data Sheet pocket or binder is kept at or near the cabinet — every chemical's SDS should be reachable in seconds, not minutes.

Eye Wash & Emergency Shower Stations



Know the location before you need it

Combination eyewash/shower units are staged near chemical storage and use areas — locate the nearest one in your department today, not during an emergency.



Flush for at least 15 minutes

For any chemical contact with eyes or skin, flush continuously and seek medical evaluation — don't stop early even if the irritation seems to ease.

Spill Response Equipment & Code Orange Response



Spill Response Equipment

WHERE TO FIND IT

Staged in or near every area that stores hazardous chemicals.

EQUIPMENT

- Chemical spill kits (absorbents, neutralizers, PPE)
- Biological spill kits (for blood/body fluids)
- Emergency eyewash stations
- Emergency safety showers



Code Orange Response

WHEN TO ACTIVATE

A spill of 500 mL or more, or any highly toxic, reactive, or explosive substance — otherwise, contact your cleaning supervisor.

STEPS

1. Isolate the area and warn nearby staff.
2. Put on the PPE specified on the SDS.
3. Contain using the correct spill kit — chemical or biological.
4. Report to your supervisor and Safety Services.

WASTE SEGREGATION

Medical Waste Management



General Waste

Non-hazardous, non-infectious waste — everyday trash generated throughout the facility.



Infectious / Biohazard

Bags, dressings, and materials contaminated with body fluids — segregated into labeled biohazard bags.



Sharps

Needles, blades, and any sharp item — into a rigid, puncture-resistant sharps container only. Never recap.



Body Fluid Spills

Blood, urine, or vomit: contain the area, disinfect per protocol using the correct PPE, and dispose of materials as biohazard waste.

Segregation colors vary by facility — always follow the labeled bins and signage in your department.

MODULE 6 OF 9

Security Management Plan

By the end of this module, you will know facility ID and access rules, and how the premises are monitored and kept secure.



ID & Access · Permits · Parking · No Smoking

WHO'S ON SITE

ID, Permits & Visitors



Wear proper ID

All staff must display facility ID at all times — it identifies you to colleagues, patients, and Security.



Contractor work permits

Contractors must hold a valid, approved work permit before beginning any work on site.



Visitor identification

All visitors sign in at reception and receive a visitor badge — report anyone in a restricted area without one.

AROUND THE FACILITY

Premises & Monitoring



CCTV coverage

Public areas, entrances, and parking are monitored — footage supports investigations when needed.



Lost & found

Found items are logged and handed to Security; report lost belongings at the Security desk.



Parking rules

Park only in designated, authorized areas — unauthorized vehicles may be relocated or reported.



No-smoking policy

Smoking is prohibited throughout the facility, including entrances and parking areas — this is also a Fire Safety requirement.

MODULE 7 OF 9

Utility Plan

By the end of this module, you will know the essential utility systems in the facility and how backup systems work.



Electrical · Backup Power · Water · IT · HVAC

Electrical Systems



Standard vs. emergency sockets

Emergency (red) sockets are backed by generator power — know which outlets in your area are which.



UPS & generators

The Uninterruptible Power Supply (UPS) covers critical equipment instantly; generators take over within seconds during a full outage.



Authorized extensions only

Only facility-approved power extensions may be used — unauthorized extensions are a fire risk and are prohibited.

BEYOND ELECTRICAL

Water, IT & HVAC



Water sources & RO plant

Potable water is treated through our Reverse Osmosis (RO) plant; know the location of manual shut-off valves.



Sewage systems

Blockages or backups should be reported immediately to Maintenance to prevent a contamination risk.



IT systems

Clinical systems run on redundant servers — report any network outage to IT immediately, especially in patient care areas.



HVAC

Heating, ventilation & air conditioning maintain air pressure and infection-control zones — never disable vents in isolation rooms.

MODULE 8 OF 9

Biomedical Plan

By the end of this module, you will understand how medical equipment is labeled and maintained, how the oxygen network works, and how to report a problem.



Equipment Labels · PPM & Inspection · Oxygen Network

EQUIPMENT YOU CAN TRUST

Equipment Labeling & Inspection



Inspection labels

Every piece of biomedical equipment carries a label showing its last inspection date and status — never use equipment past its due date.



Planned Preventive Maintenance (PPM)

Equipment is serviced on a fixed schedule by Biomedical Engineering — report anything overdue or missing its label.



Out-of-service tagging

Faulty equipment gets tagged "Out of Service" and removed from use immediately — never attempt to repair it yourself.



EQUIPMENT RELOCATION

Never relocate medical equipment on your own.

Every approved location has been specifically assessed for the equipment placed there. Moving equipment between locations without authorization — especially radiation and laser equipment (see Module 9) — bypasses that assessment and can create serious, unassessed risk. Always coordinate with Biomedical Engineering before any equipment is moved, even temporarily.

A CRITICAL RESOURCE

Oxygen Cylinders & Network



Piped oxygen network

Central oxygen supplies patient areas through wall outlets — know the location of your zone's emergency shut-off valve.



Cylinder storage

Store cylinders upright, secured with a chain or stand, and clearly separated between "full" and "empty."



Keep sources of ignition away

Oxygen is an oxidizer — never store or use cylinders near open flame, sparks, or smoking materials (see Module 4).

SHARED RESPONSIBILITY

Everyone's Role



Spot it

If equipment looks damaged, or its inspection label is missing or expired, stop using it.



Report it

Notify Biomedical Engineering immediately — do not attempt your own repair or bypass a fault.

MODULE 9 OF 9

Radiation & Laser Safety

By the end of this module, you will understand ionizing vs. non-ionizing radiation, the ALARA principles and dose limits, laser classifications and hazards, and who to contact for either.



Ionizing & Non-Ionizing Radiation · ALARA · Dose Limits · Laser Safety

Ionizing vs. Non-Ionizing



Ionizing Radiation

Carries enough energy to remove electrons from atoms — this is the kind that requires dosimeters, shielding, and dose tracking. Examples: X-rays, CT.



Non-Ionizing Radiation

Lower-energy radiation that doesn't ionize atoms — still requires precautions, just different ones. Examples: MRI, Ultrasound, and laser light.

Radiation is the emission or transmission of energy as waves or particles through space or a material medium.

HIGH-RISK AREAS

Radiation Areas & Precautions



MRI safety

The MRI magnet is always on — and it's a non-ionizing magnetic hazard, not radiation exposure. Metal objects, oxygen cylinders, and unscreened staff must never enter the MRI room — screening applies every time, no exceptions.



X-ray & imaging areas (ionizing)

Lead aprons, shielding, and posted warning signs are mandatory whenever imaging equipment is in use — stand behind the shield or leave the room.

Principles of Radiation Protection



Justification

Any activity that exposes people to radiation must have benefits that clearly justify the risk.



Optimization

Exposure is kept to the minimum reasonable level — this is the ALARA principle in practice.



Limitation

A firm upper dose limit applies to every worker and member of the public, outside of medical exposures.

As Low As Reasonably Achievable



Time

Minimize the time you spend near a radiation source — less time means less cumulative dose.



Distance

Radiation intensity drops sharply with distance — step back whenever you don't need to be close.



Shielding

Lead aprons, barriers, and walls absorb radiation before it reaches you.

STAYING ACCOUNTABLE

Monitoring & Responsibility



Personal dosimeters (OSL)

Staff who work in or near radiation areas must wear an Optically Stimulated Luminescence (OSL) dosimeter badge at all times to track exposure.



Wear your lead apron

A lead apron is required PPE whenever you're in the room during exposure — it's not optional, even for a quick scan.



Radiation Safety Officer (RSO)

The RSO oversees compliance, monitors exposure records, and is your point of contact for any radiation safety concern.

KNOW YOUR LIMITS

Dose Limits

Personal dose should never exceed 2 mSv within any 3-month period, or 20 mSv within a year.



Reading Level

1.25 mSv

No action needed — this is expected background exposure.



Notification Level

1.25 – 3.75 mSv

Inform the employee and place them under monitoring.



Investigation Level

Above 3.75 mSv

Follow the facility's over-exposure investigation policy.

What Is a Laser?



Light Amplification by Stimulated Emission of Radiation

That's what "LASER" stands for. Like MRI and ultrasound, laser light is a form of non-ionizing radiation — but it carries its own distinct hazards.

KNOW THE POWER LEVEL

Laser Classification

Higher classes mean greater hazard — check the label before you work near any laser device.

Class 1 & 1M

Safe under normal use conditions

Class 2 & 2M

Low power — the blink reflex offers some protection

Class 3

Moderate power — direct viewing is hazardous

Class 3R

Low risk if safety precautions are followed

Class 3B

Hazardous — direct and reflected viewing

Class 4

Severe hazard — eyes, skin, and fire risk

WHAT CAN GO WRONG

Laser Hazards



Eyes — cataracts

Laser exposure can cause permanent eye damage, including cataracts, often without immediate pain to warn you.



Skin — burns

Direct or reflected beams can cause skin burns, especially with higher-class lasers.



Electrical — can be lethal

The high-voltage components inside laser equipment carry a separate, serious electrical hazard.

KNOW THE TRIGGERS

Common Sources of Laser Exposure



Accidental Eye Exposure



Misaligned Laser Beam



Lack of Eye Protection



Equipment Malfunction

PREVENT THE EXPOSURE

Key Laser Safety Measures



Eye Protection



Beam Management



Control Reflections



Environmental Safety



Training



Program Complete

You've completed all nine modules of Facility Management & Safety. This knowledge is only useful if it's current — revisit this program whenever plans, codes, or procedures are updated.

- Complete the program assessment / quiz
- Sign the orientation acknowledgment form
- Locate your unit's evacuation route and assembly point
- Save key emergency numbers to your phone

Questions? Contact your Safety & Security Manager